

# Spark Preparation

We check if we are in Google Colab. If this is the case, install all necessary packages.

To run spark in Colab, we need to first install all the dependencies in Colab environment i.e. Apache Spark 3.3.2 with hadoop 3.3, Java 8 and Findspark to locate the spark in the system. The tools installation can be carried out inside the Jupyter Notebook of the Colab.

Learn more from [A Must-Read Guide on How to Work with PySpark on Google Colab for Data Scientists!](#)

```
In [ ]: try:
        import google.colab
        IN_COLAB = True
    except:
        IN_COLAB = False
```

```
In [ ]: if IN_COLAB:
        !apt-get install openjdk-8-jdk-headless -qq > /dev/null
        !wget -q https://dlcdn.apache.org/spark/spark-3.3.2/spark-3.3.2-bin-hadoop3.tgz
        !tar xf spark-3.3.2-bin-hadoop3.tgz
        !mv spark-3.3.2-bin-hadoop3 spark
        !pip install -q findspark
        import os
        os.environ["JAVA_HOME"] = "/usr/lib/jvm/java-8-openjdk-amd64"
        os.environ["SPARK_HOME"] = "/content/spark"
```

## Start a Local Cluster

```
In [1]: from pyspark.sql import SparkSession
        from pyspark.sql.functions import avg, max
```

```
In [2]: spark_url = 'local'
        spark = SparkSession.builder\
            .master(spark_url)\
            .appName('Spark assignment')\
```

```
.config('spark.ui.port', '4040')\  
.getOrCreate()
```

```
WARNING: Using incubator modules: jdk.incubator.vector  
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties  
26/04/01 15:49:41 WARN Utils: Your hostname, Chatrins-MacBook-Air.local, resolves to a loopback address: 127.0.0.1;  
using 192.168.105.1 instead (on interface bridge100)  
26/04/01 15:49:41 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
26/04/01 15:49:42 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-jav  
a classes where applicable
```

## Spark Assignment

Based on the movie review dataset in 'netflix-rotten-tomatoes-metacritic-imdb.csv', answer the below questions.

**Note:** do not clean or remove missing data

```
In [3]: df = spark.read.csv('netflix-rotten-tomatoes-metacritic-imdb.csv', header=True, inferSchema=True)
```

```
In [4]: df.show(5)
```

|                                | Title                | Genre                              | Tags                               | Languages        | Series or Movie | Hidden Gem Score     | Cou          |
|--------------------------------|----------------------|------------------------------------|------------------------------------|------------------|-----------------|----------------------|--------------|
| ntry Availability              | Runtime              | Director                           | Writer                             | Actors           | View Rating     | IMDb Score           | Rott         |
| en Tomatoes Score              | Metacritic Score     | Awards Received                    | Awards Nominated For               | Boxoffice        | Release Date    | Netflix Release Date |              |
|                                | Production House     | Netflix Link                       | IMDb Link                          | Summary          | IMDb Votes      |                      | Image        |
|                                | Poster               | TMDb Trailer                       | Trailer Site                       |                  |                 |                      |              |
| Lets Fight Ghost               | Crime, Drama, Fan... | Comedy Programmes...               | Swedish, Spanish                   |                  | Series          | 4.3                  |              |
| Thailand < 30 minutes          | Tomas Alfredson      | John Ajvide Lindq...               | Kåre Hedebrant, P...               |                  | R               | 7.9                  |              |
| 98.0                           | 82.0                 | 74.0                               | 57.0 \$2,122,065                   | 12 Dec 2008      |                 | 2021-03-04           | Canal+, Sand |
| rew M... https://www.netfl...  | https://www.imdb.... | A med student wit...               | 205926.0 https://occ-0-470...      | https://m.me     |                 |                      |              |
| dia-a...                       | NULL                 | NULL                               |                                    |                  |                 |                      |              |
| HOW TO BUILD A GIRL            | Comedy               | Dramas,Comedies,F...               | English                            |                  | Movie           | 7.0                  |              |
| Canada                         | 1-2 hour             | Coky Giedroyc                      | Caitlin Moran Paddy Considine, ... |                  | R               | 5.8                  |              |
| 79.0                           | 69.0                 | 1.0                                | NULL  \$70,632                     | 08 May 2020      |                 | 2021-03-04           | Film 4, Monu |
| menta... https://www.netfl...  | https://www.imdb.... | When nerdy Johann...               | 2838.0 https://occ-0-108...        | https://m.me     |                 |                      |              |
| dia-a... https://www.youtu...  | YouTube              |                                    |                                    |                  |                 |                      |              |
| Centigrade                     | Drama, Thriller      | Thrillers                          | English                            |                  | Movie           | 6.4                  |              |
| Canada                         | 1-2 hour             | Brendan Walsh Brendan Walsh, Da... | Genesis Rodriguez...               | Unrated          | 4.3             |                      |              |
| NULL                           | 46.0                 | NULL                               | NULL  \$16,263                     | 28 Aug 2020      |                 | 2021-03-04           |              |
| NULL https://www.netfl...      | https://www.imdb.... | Trapped in a froz...               | 1720.0 https://occ-0-108...        | https://m.media- |                 |                      |              |
| a... https://www.youtu...      | YouTube              |                                    |                                    |                  |                 |                      |              |
| ANNE+                          | Drama                | TV Dramas,Romanti...               | Turkish                            |                  | Series          | 7.7                  | Be           |
| lgium,Netherlands < 30 minutes |                      | NULL                               | NULL Vahide Perçin, Go...          |                  | NULL            | 6.5                  |              |
| NULL                           | NULL                 | 1.0                                | NULL  NULL                         | 01 Oct 2016      |                 | 2021-03-04           |              |
| NULL https://www.netfl...      | https://www.imdb.... | Upon moving into ...               | 1147.0 https://occ-0-148...        | https://m.media- |                 |                      |              |
| a...                           | NULL                 | NULL                               |                                    |                  |                 |                      |              |
| Moxie                          | Animation, Short,... | Social Issue Dram...               | English                            |                  | Movie           | 8.1                  | Lit          |
| huania,Poland,...              | 1-2 hour             | Stephen Irwin                      | NULL                               | Ragga Gudrun     | NULL            | 6.3                  |              |
| NULL                           | NULL                 | NULL                               | 4.0  NULL                          | 22 Sep 2011      |                 | 2021-03-04           |              |
| NULL https://www.netfl...      | https://www.imdb.... | Inspired by her m...               | 63.0 https://occ-0-403...          | https://m.media- |                 |                      |              |
| a...                           | NULL                 | NULL                               |                                    |                  |                 |                      |              |

```

-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+

```

only showing top 5 rows

26/04/01 15:49:46 WARN SparkStringUtils: Truncated the string representation of a plan since it was too large. This behavior can be adjusted by setting 'spark.sql.debug.maxToStringFields'.

## What is the maximum and average of the overall hidden gem score?

```
In [5]: df.select(max(df['Hidden Gem Score']), avg(df['Hidden Gem Score'])).show()
```

```

+-----+-----+
|max(Hidden Gem Score)|avg(Hidden Gem Score)|
+-----+-----+
|                9.8|    5.937551386501226|
+-----+-----+

```

## How many movies that are available in Korea?

```
In [6]: from pyspark.sql.functions import split, explode
```

```
In [7]: df_exploded = df.withColumn("Country", explode(split(df['Country Availability'], ',')))
```

```
In [8]: country_count = df_exploded.groupBy('Country').count().sort("count", ascending=False)
country_count.show()
```

| Country        | count |
|----------------|-------|
| United Kingdom | 6311  |
| Czech Republic | 6237  |
| Japan          | 6115  |
| Hungary        | 5989  |
| Canada         | 5930  |
| Singapore      | 5876  |
| Thailand       | 5876  |
| India          | 5826  |
| Slovakia       | 5820  |
| Australia      | 5820  |
| Romania        | 5759  |
| United States  | 5718  |
| South Africa   | 5553  |
| Lithuania      | 5541  |
| Germany        | 5539  |
| Russia         | 5491  |
| Switzerland    | 5472  |
| Belgium        | 5437  |
| Malaysia       | 5394  |
| Iceland        | 5364  |

only showing top 20 rows

```
In [9]: country_count[country_count['Country']=='South Korea'].show()
```

| Country     | count |
|-------------|-------|
| South Korea | 4845  |

```
In [34]: lang_exploded = df.withColumn("lang", explode(split(df['Languages'], ', ')))
lang_count = lang_exploded.groupBy('lang').count().sort('count', ascending=False)
lang_count[lang_count['lang']=='Korean'].show()
```

```
+-----+-----+
| lang | count |
+-----+-----+
| Korean | 735 |
+-----+-----+
```

## Which director has the highest average hidden gem score?

```
In [10]: df.groupBy('Director').avg("Hidden Gem Score",).sort("avg(Hidden Gem Score)", ascending=False).show(1)
```

```
+-----+-----+
| Director | avg(Hidden Gem Score) |
+-----+-----+
| Dorin Marcu | 9.8 |
+-----+-----+
```

only showing top 1 row

## How many genres are there in the dataset?

```
In [21]: explode_genre_df = df.withColumn("genre", explode(split(df["Genre"], ', ')))
explode_genre_df.show()
```

|                                                                               | Title                                                                                   | genre                                                                    | Tags                                                                                                                | Languages            | Series or Movie | Hidden Gem Score     | Country Av  |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------|-----------------|----------------------|-------------|
| ailability                                                                    | Runtime                                                                                 | Director                                                                 | Writer                                                                                                              | Actors               | View Rating     | IMDb Score           | Rotten Toma |
| toes Score                                                                    | Metacritic Score                                                                        | Awards Received                                                          | Awards Nominated For                                                                                                | Boxoffice            | Release Date    | Netflix Release Date | Pr          |
| oduction House                                                                | Netflix Link                                                                            | IMDb Link                                                                | Summary                                                                                                             | IMDb Votes           | Image           |                      |             |
| Poster                                                                        | TMDb Trailer                                                                            | Trailer Site                                                             |                                                                                                                     |                      |                 |                      |             |
| Lets Fight Ghost  Crime Comedy Programmes...  Swedish, Spanish  Series  4.3   | Thailand < 30 minutes Tomas Alfredson John Ajvide Lindq... Kåre Hedebrant, P...  R  7.9 | 98.0  82.0  74.0  57.0 \$2,122,065  12 Dec 2008  2021-03-04 Canal+, Sand | rew M... https://www.netfl... https://www.imdb.... A med student wit...  205926.0 https://occ-0-470... https://m.me | dia-a...  NULL  NULL |                 |                      |             |
| Lets Fight Ghost  Drama Comedy Programmes...  Swedish, Spanish  Series  4.3   | Thailand < 30 minutes Tomas Alfredson John Ajvide Lindq... Kåre Hedebrant, P...  R  7.9 | 98.0  82.0  74.0  57.0 \$2,122,065  12 Dec 2008  2021-03-04 Canal+, Sand | rew M... https://www.netfl... https://www.imdb.... A med student wit...  205926.0 https://occ-0-470... https://m.me | dia-a...  NULL  NULL |                 |                      |             |
| Lets Fight Ghost  Fantasy Comedy Programmes...  Swedish, Spanish  Series  4.3 | Thailand < 30 minutes Tomas Alfredson John Ajvide Lindq... Kåre Hedebrant, P...  R  7.9 | 98.0  82.0  74.0  57.0 \$2,122,065  12 Dec 2008  2021-03-04 Canal+, Sand | rew M... https://www.netfl... https://www.imdb.... A med student wit...  205926.0 https://occ-0-470... https://m.me | dia-a...  NULL  NULL |                 |                      |             |
| Lets Fight Ghost  Horror Comedy Programmes...  Swedish, Spanish  Series  4.3  | Thailand < 30 minutes Tomas Alfredson John Ajvide Lindq... Kåre Hedebrant, P...  R  7.9 | 98.0  82.0  74.0  57.0 \$2,122,065  12 Dec 2008  2021-03-04 Canal+, Sand | rew M... https://www.netfl... https://www.imdb.... A med student wit...  205926.0 https://occ-0-470... https://m.me | dia-a...  NULL  NULL |                 |                      |             |
| Lets Fight Ghost  Romance Comedy Programmes...  Swedish, Spanish  Series  4.3 | Thailand < 30 minutes Tomas Alfredson John Ajvide Lindq... Kåre Hedebrant, P...  R  7.9 | 98.0  82.0  74.0  57.0 \$2,122,065  12 Dec 2008  2021-03-04 Canal+, Sand | rew M... https://www.netfl... https://www.imdb.... A med student wit...  205926.0 https://occ-0-470... https://m.me | dia-a...  NULL  NULL |                 |                      |             |
| HOW TO BUILD A GIRL  Comedy Dramas,Comedies,F...  English  Movie  7.0         |                                                                                         |                                                                          |                                                                                                                     |                      |                 |                      |             |

|            |                      |                                    |                                    |                       |                                       |
|------------|----------------------|------------------------------------|------------------------------------|-----------------------|---------------------------------------|
| Canada     | 1-2 hour             | Coky Giedroyc                      | Caitlin Moran Paddy Considine, ... | R                     | 5.8                                   |
| 79.0       | 69.0                 | 1.0                                | NULL                               | \$70,632  08 May 2020 | 2021-03-04 Film 4, Monu               |
| menta...   | https://www.netfl... | https://www.imdb....               | When nerdy Johann...               | 2838.0                | https://occ-0-108... https://m.me     |
| dia-a...   | https://www.youtu... | YouTube                            |                                    |                       |                                       |
|            | Centigrade           | Drama                              | Thrillers                          | English               | Movie                                 |
| Canada     | 1-2 hour             | Brendan Walsh Brendan Walsh, Da... | Genesis Rodriguez...               | Unrated               | 4.3                                   |
| NULL       | 46.0                 | NULL                               | NULL                               | \$16,263  28 Aug 2020 | 2021-03-04                            |
| NULL       | https://www.netfl... | https://www.imdb....               | Trapped in a froz...               | 1720.0                | https://occ-0-108... https://m.media- |
| a...       | https://www.youtu... | YouTube                            |                                    |                       |                                       |
|            | Centigrade           | Thriller                           | Thrillers                          | English               | Movie                                 |
| Canada     | 1-2 hour             | Brendan Walsh Brendan Walsh, Da... | Genesis Rodriguez...               | Unrated               | 4.3                                   |
| NULL       | 46.0                 | NULL                               | NULL                               | \$16,263  28 Aug 2020 | 2021-03-04                            |
| NULL       | https://www.netfl... | https://www.imdb....               | Trapped in a froz...               | 1720.0                | https://occ-0-108... https://m.media- |
| a...       | https://www.youtu... | YouTube                            |                                    |                       |                                       |
|            | ANNE+                | Drama TV Dramas,Romanti...         |                                    | Turkish               | Series                                |
| etherlands | < 30 minutes         | NULL                               | NULL                               | Vahide Perçin, Go...  | NULL                                  |
| NULL       | NULL                 | 1.0                                | NULL                               | NULL                  | 01 Oct 2016                           |
| NULL       | https://www.netfl... | https://www.imdb....               | Upon moving into ...               | 1147.0                | https://occ-0-148... https://m.media- |
| a...       | NULL                 | NULL                               |                                    |                       |                                       |
|            | Moxie                | Animation Social Issue Dram...     |                                    | English               | Movie                                 |
| Poland,... | 1-2 hour             | Stephen Irwin                      | NULL                               | Ragga Gudrun          | NULL                                  |
| NULL       | NULL                 | NULL                               | 4.0                                | NULL                  | 22 Sep 2011                           |
| NULL       | https://www.netfl... | https://www.imdb....               | Inspired by her m...               | 63.0                  | https://occ-0-403... https://m.media- |
| a...       | NULL                 | NULL                               |                                    |                       |                                       |
|            | Moxie                | Short Social Issue Dram...         |                                    | English               | Movie                                 |
| Poland,... | 1-2 hour             | Stephen Irwin                      | NULL                               | Ragga Gudrun          | NULL                                  |
| NULL       | NULL                 | NULL                               | 4.0                                | NULL                  | 22 Sep 2011                           |
| NULL       | https://www.netfl... | https://www.imdb....               | Inspired by her m...               | 63.0                  | https://occ-0-403... https://m.media- |
| a...       | NULL                 | NULL                               |                                    |                       |                                       |
|            | Moxie                | Drama Social Issue Dram...         |                                    | English               | Movie                                 |
| Poland,... | 1-2 hour             | Stephen Irwin                      | NULL                               | Ragga Gudrun          | NULL                                  |
| NULL       | NULL                 | NULL                               | 4.0                                | NULL                  | 22 Sep 2011                           |
| NULL       | https://www.netfl... | https://www.imdb....               | Inspired by her m...               | 63.0                  | https://occ-0-403... https://m.media- |
| a...       | NULL                 | NULL                               |                                    |                       |                                       |
|            | The Con-Heartist     | Comedy Romantic Comedies...        |                                    | Thai                  | Movie                                 |
| Thailand   | > 2 hrs              | Mez Tharatorn Pattaranad Bhiboo... | Thiti Mahayotaruk...               | NULL                  | 7.4                                   |
| NULL       | NULL                 | NULL                               | NULL                               | NULL                  | 03 Dec 2020                           |
| NULL       | https://www.netfl... | https://www.imdb....               | After her ex-boyf...               | 131.0                 | https://occ-0-218... https://m.media- |
| a...       | https://www.youtu... | YouTube                            |                                    |                       |                                       |
|            | The Con-Heartist     | Romance Romantic Comedies...       |                                    | Thai                  | Movie                                 |
| Thailand   | > 2 hrs              | Mez Tharatorn Pattaranad Bhiboo... | Thiti Mahayotaruk...               | NULL                  | 7.4                                   |

|                           |                      |                              |                             |                      |             |            |
|---------------------------|----------------------|------------------------------|-----------------------------|----------------------|-------------|------------|
| NULL                      | NULL                 | NULL                         | NULL                        | NULL                 | 03 Dec 2020 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | After her ex-boyf...         | 131.0 https://occ-0-218...  | https://m.media-     |             |            |
| a... https://www.youtu... | YouTube              |                              |                             |                      |             |            |
|                           | Gleboka woda         | Drama TV Dramas, Polish ...  | Polish                      | Series               | 8.7         |            |
| Poland < 30 minutes       | NULL                 | NULL                         | NULL                        | Marcin Dorocinski... | NULL        | 7.5        |
| NULL                      | NULL                 | 2.0                          | 4.0                         | NULL                 | 14 Jun 2011 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | A group of social...         | 47.0 https://occ-0-250...   | https://m.media-     |             |            |
| a...                      | NULL                 | NULL                         |                             |                      |             |            |
|                           | Instynkt             | Crime TV Dramas, Crime T...  | Polish                      | Series               | 6.9         |            |
| Poland < 30 minutes       | NULL                 | NULL                         | NULL                        | Paweł Królikowski... | NULL        | 3.9        |
| NULL                      | NULL                 | NULL                         | NULL                        | NULL                 | 03 Mar 2011 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | An enigmatic comm...         | 107.0 https://occ-0-250...  | https://m.media-     |             |            |
| a...                      | NULL                 | NULL                         |                             |                      |             |            |
|                           | Only a Mother        | Drama Social Issue Dram...   | Swedish                     | Movie                | 8.3         | Lithuania, |
| Poland,...                | 1-2 hour             | Alf Sjöberg                  | Ivar Lo-Johansson           | Ulf Palme, Ragnar... | NULL        | 6.7        |
| NULL                      | NULL                 | 2.0                          | 1.0                         | NULL                 | 31 Oct 1949 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | An unhappily marr...         | 88.0 https://occ-0-285...   | https://m.media-     |             |            |
| a...                      | NULL                 | NULL                         |                             |                      |             |            |
|                           | Snowroller           | Comedy Sports Movies, Spo... | Swedish, English,...        | Movie                | 5.3         | Lithuania, |
| Poland,...                | 1-2 hour             | Lasse Åberg                  | Lasse Åberg, Bo J...        | Jon Skolmen, Ceci... | NULL        | 6.6        |
| NULL                      | NULL                 | NULL                         | NULL                        | NULL                 | 04 Oct 1985 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | Two friends take ...         | 5926.0 https://occ-0-285... | https://m.media-     |             |            |
| a...                      | NULL                 | NULL                         |                             |                      |             |            |
|                           | Sunes Summer         | Comedy Children & Family...  | Swedish                     | Movie                | 7.5         | Lithuania, |
| Poland,...                | 1-2 hour             | Jon Holmberg                 | Jon Holmberg, Dan...        | Elis Gerdt, Tea S... | NULL        | 5.5        |
| NULL                      | NULL                 | NULL                         | 7.0                         | NULL                 | 30 Nov 2018 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | A family camping ...         | 560.0 https://occ-0-285...  | https://m.media-     |             |            |
| a...                      | NULL                 | NULL                         |                             |                      |             |            |
|                           | Sunes Summer         | Family Children & Family...  | Swedish                     | Movie                | 7.5         | Lithuania, |
| Poland,...                | 1-2 hour             | Jon Holmberg                 | Jon Holmberg, Dan...        | Elis Gerdt, Tea S... | NULL        | 5.5        |
| NULL                      | NULL                 | NULL                         | 7.0                         | NULL                 | 30 Nov 2018 | 2021-03-03 |
| NULL https://www.netfl... | https://www.imdb.... | A family camping ...         | 560.0 https://occ-0-285...  | https://m.media-     |             |            |
| a...                      | NULL                 | NULL                         |                             |                      |             |            |

```

+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+-----+

```

only showing top 20 rows

```
In [26]: genre_count = explode_genre_df.groupby('genre').count()  
genre_count.show()
```

```
+-----+-----+  
|      genre|count|  
+-----+-----+  
|      Crime| 1932|  
|    Romance| 2445|  
|    Thriller| 2739|  
|  Adventure| 1809|  
|      Drama| 6359|  
|        War|   330|  
|Documentary| 1030|  
| Reality-TV|   191|  
|      Family| 1433|  
|      Fantasy| 1594|  
| Game-Show|    52|  
|       Adult|    15|  
|      History|   527|  
|      Mystery| 1190|  
|      Musical|   228|  
| Animation| 1665|  
|       Music|   426|  
| Film-Noir|     2|  
|      Horror| 1070|  
|       Short|   422|  
+-----+-----+
```

only showing top 20 rows

```
In [27]: genre_count.count()
```

```
Out[27]: 28
```